

Sentiment Classification on Amazon Product Reviews: A Comparative Study of Traditional Machine Learning Approaches

Hanrong Zhang

Introduction

Sentiment analysis — the computational task of identifying and categorizing opinions expressed in text has become one of the most actively studied problems in natural language processing. As e-commerce platforms have grown to host hundreds of millions of user-generated reviews, the ability to automatically understand the sentiment embedded in these reviews carries significant practical value: businesses can monitor product reception, detect quality issues, and understand customer needs at scale without manual reading.

Prior work has established a clear taxonomy of approaches to this problem. Devika, Sunitha, and Ganesh (2016) organize the field around three broad method categories — machine learning, rule-based, and lexicon-based approaches — and compare individual classifiers including Support Vector Machines, Naive Bayes, and N-gram models across document-level, sentence-level, and aspect-level tasks. Their analysis concludes that machine learning methods generally yield the best performance, while also noting that classifiers trained in one domain often generalize poorly to others. Birjali, Kasri, and Beni-Hssane (2021) extend this survey to cover deep learning architectures including RNN, LSTM, and transformer-based models such as BERT, situating these within the current state of the art. Crucially, their treatment of feature extraction techniques — including TF-IDF, bag-of-words, and contextual word embeddings — provides a principled basis for understanding where TF-IDF sits in the landscape of text representations: strong and interpretable for structured review data, but limited in its ability to capture word order or mixed-sentiment context. Both surveys directly inform the method choices and evaluation strategy of this project.

The Amazon product review corpus is among the most widely used datasets in this domain. Hou et al. (2024) introduced the Amazon Reviews 2023 dataset, a large-scale collection of over 571 million reviews spanning 33 product categories, collected through September 2023. This dataset is notable for its size, temporal depth, and richness of metadata, including fine-grained timestamps and helpfulness votes, making it an ideal benchmark for sentiment classification research.

The central challenge this project addresses is how effectively traditional machine learning methods — Complement Naive Bayes, Logistic Regression, and Linear SVM — can predict sentiment from raw review text when combined with TF-IDF feature extraction. Beyond the

standard binary positive/negative framing common in prior work, this project introduces a more nuanced three-class formulation that explicitly models neutral sentiment (3-star reviews). Neutral reviews occupy a semantically ambiguous space: they often contain mixed praise and criticism, and their linguistic patterns overlap heavily with both positive and negative language. This ambiguity makes them particularly difficult for bag-of-words approaches to classify correctly, and the analysis of this failure is a central contribution of the project. The project also deliberately restricts itself to traditional machine learning with TF-IDF features — rather than adopting transformer-based models as advocated by Birjali et al. (2021) — in order to build a complete, interpretable end-to-end pipeline and establish a clear baseline against which more complex future approaches can be compared.

Methods

Dataset and Preprocessing

This project uses the All Beauty category from the Amazon Reviews 2023 dataset, accessed via the Hugging Face `datasets` library (Hou et al., 2024). The full category contains 701,528 reviews. To enable manageable experimentation, 50,000 reviews were sampled uniformly at random using a fixed random seed (42) for reproducibility. Reviews with missing or empty text fields were removed prior to sampling.

Two labeling schemes were implemented. For binary classification, reviews with ratings of 4 or 5 were labeled as Positive (1) and reviews with ratings of 1, 2, or 3 were labeled as Negative (0). For the ternary classification experiment, ratings 1–2 were labeled Negative (0), rating 3 was labeled Neutral (1), and ratings 4–5 were labeled Positive (2). This separation isolates the semantically ambiguous middle ground for explicit analysis.

Text preprocessing was applied uniformly across both experiments. Each review was lowercased, stripped of HTML tags using regular expressions, reduced to alphabetic characters only, and normalized for whitespace. Feature extraction was performed using scikit-learn's `TfidfVectorizer` (Pedregosa et al., 2011) with the following configuration: a maximum vocabulary size of 50,000 features, unigram and bigram token ranges, sublinear TF scaling, a minimum document frequency of 3, and standard English stop word removal. Each dataset was split into 80% training and 20% test sets using stratified sampling to preserve class proportions.

Models

Three classifiers were evaluated in both experiments. Complement Naive Bayes (ComplementNB) is a variant of Multinomial Naive Bayes designed to address class imbalance by estimating parameters from the complement of each class (Rennie et al., 2003). Logistic Regression was configured with a maximum of 1,000 iterations and regularization strength $C=1.0$. Linear SVM (LinearSVC) was configured with a maximum of 2,000 iterations and $C=1.0$. Each classifier was wrapped in a scikit-learn Pipeline together with the TF-IDF vectorizer, ensuring that the vectorizer was fit only on training data to prevent data leakage.

Class Imbalance Mitigation

In the ternary setting, the Neutral class constitutes only approximately 8% of the sampled data (3,983 out of 50,000 reviews). To address this imbalance, a second set of models was trained using `class_weight='balanced'`, which adjusts sample weights inversely proportional to class frequencies during training. This was applied to Logistic Regression and LinearSVC. ComplementNB does not support the `class_weight` parameter in scikit-learn and was therefore left unchanged, serving as an informative contrast.

Evaluation Metrics

Model performance was assessed using overall accuracy, per-class precision, recall, and F1-score, macro-averaged F1, and confusion matrices. For the ternary experiment, confusion matrices were normalized by true class to show row-wise recall rates. ROC curves and AUC scores were computed for the Logistic Regression model under the one-vs-rest scheme. An error analysis module was also implemented to examine the distribution of misclassification directions for Neutral reviews and to compare average word counts between correctly and incorrectly classified neutral samples.

Results

Binary Classification

The binary experiment demonstrated that all three models achieve strong and closely comparable performance. Logistic Regression achieved the highest accuracy at 87.65%, followed by Linear SVM at 87.48% and Complement Naive Bayes at 87.20%. Figure 1 summarizes these results.

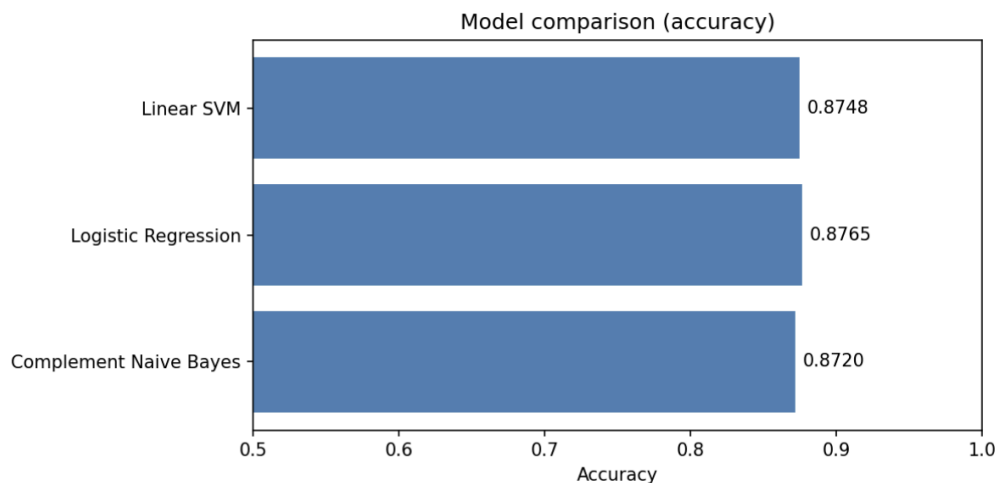


Figure 1. Accuracy comparison of three classifiers on the binary sentiment classification task.

As shown in Figure 2, per-class performance reveals a consistent asymmetry across all three models: Positive reviews are classified with substantially higher recall than Negative reviews. This pattern reflects the class imbalance in the binary setting — approximately 71% of samples are Positive — which biases all models toward predicting the majority class.

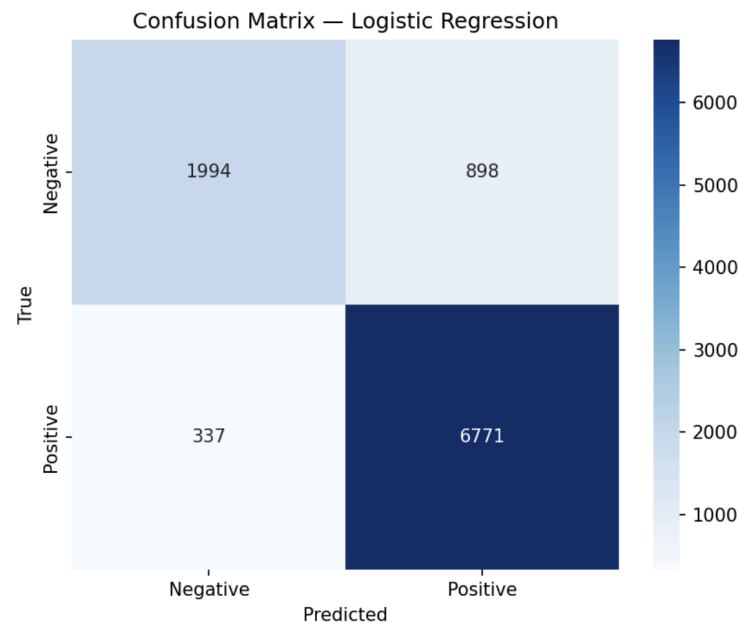


Figure 2. Confusion matrix for Logistic Regression on the binary classification task.

Ternary Classification

Introducing the Neutral class as a third category revealed a fundamental limitation of TF-IDF-based models. The class distribution, shown in Figure 3, highlights the degree of imbalance: Positive reviews account for approximately 71% of the sample, Negative for 21%, and Neutral for only 8% (3,983 samples).

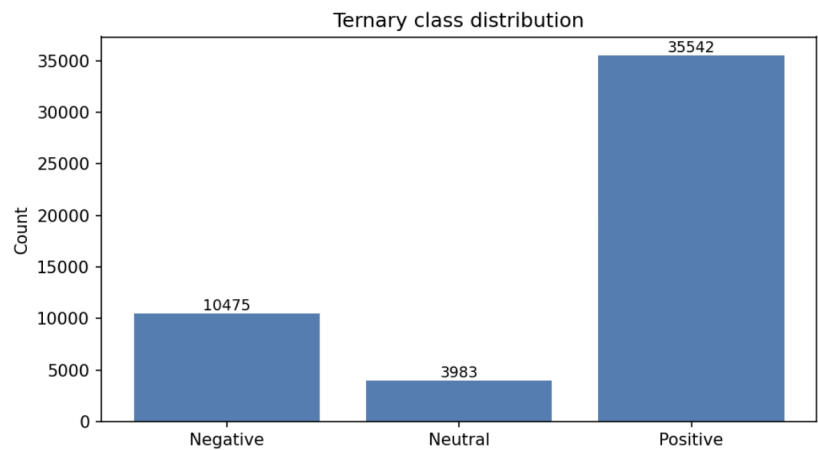


Figure 3. Class distribution in the ternary sentiment classification dataset (50,000 samples).

Overall accuracy remained relatively high across all three models, but this masks a near-complete failure on the Neutral class. Neutral F1 scores were 0.116 for Complement Naive Bayes, 0.110 for Logistic Regression, and 0.179 for Linear SVM. The divergence between accuracy and macro F1 is itself a meaningful finding: because Neutral reviews represent only 8% of test samples, a model can achieve high accuracy while almost completely failing to identify them. When evaluated by macro-averaged F1 — which weights all classes equally regardless of size — Linear SVM ranked first among the three models, despite not having the highest overall accuracy. This underscores the inadequacy of accuracy as a sole evaluation metric in imbalanced settings.

The normalized confusion matrix for Logistic Regression, shown in Figure 4, reveals the directional asymmetry in Neutral misclassification: the majority of true Neutral reviews are predicted as Positive rather than Negative, reflecting the model's strong prior toward the dominant class

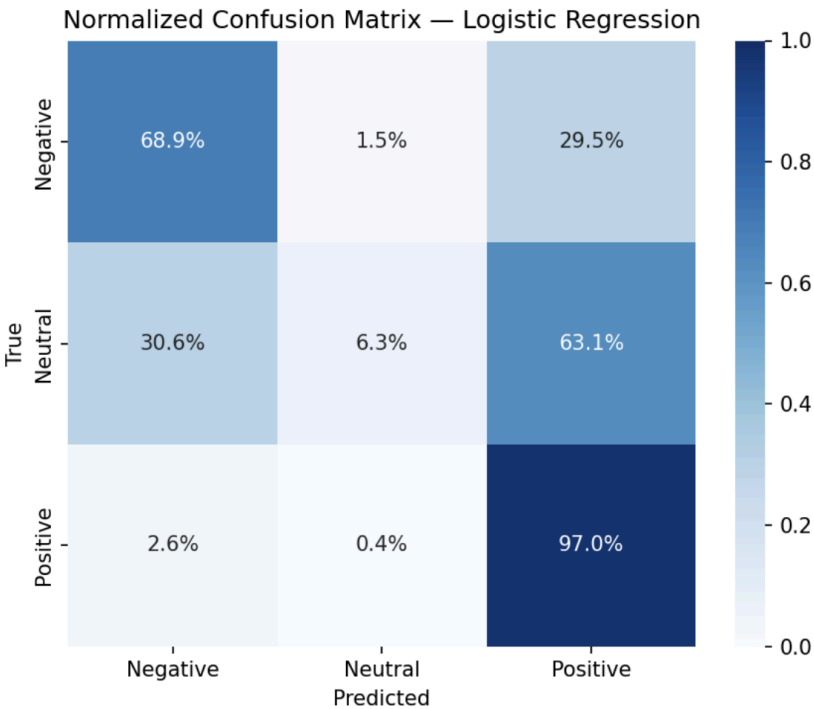


Figure 4. Normalized confusion matrix for Logistic Regression on the ternary classification task. Values represent the proportion of true-class samples predicted as each label.

ROC analysis for Logistic Regression, shown in Figure 5, confirms the Neutral class difficulty quantitatively: the Neutral AUC is visibly lower than those of the Negative and Positive classes, indicating that the model has substantially less discriminative power for this category.

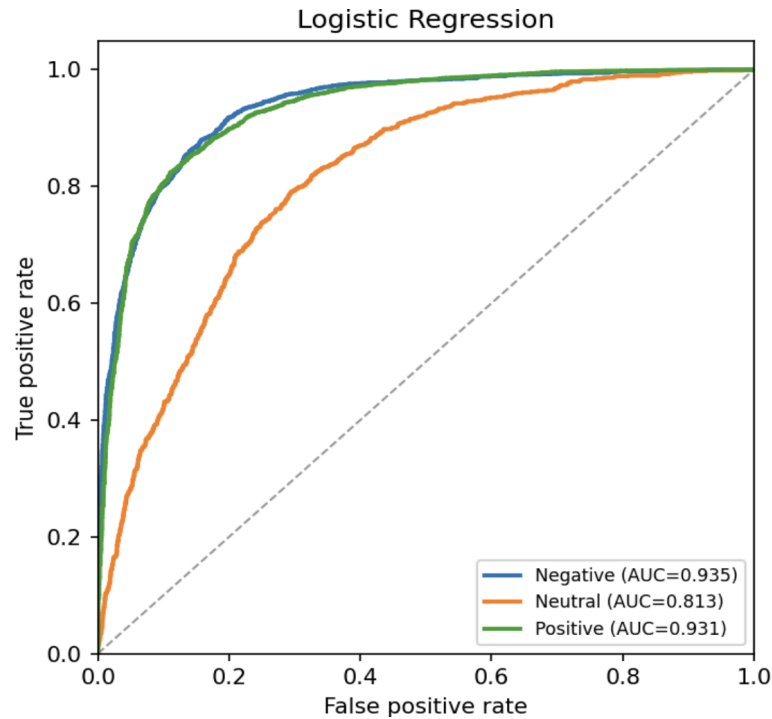


Figure 5. One-vs-rest ROC curves for Logistic Regression on the ternary classification task.

Error Analysis

Qualitative examination of misclassified Neutral reviews revealed two distinct failure modes. Reviews predicted as Positive — such as "Makes great curls but hard to piece hair in sometimes to not tangle. Charges easy but didn't last long enough to complete a full head" — contain functional product language alongside moderate criticism. Reviews predicted as Negative — such as "They look weirdly small... there is a very strong odor to them" — contain complaint language that superficially resembles lower-rated reviews. In both cases, the mixed sentiment structure cannot be resolved by TF-IDF features that treat each token independently. Additionally, correctly classified Neutral reviews had a lower average word count (31.78 words) than misclassified ones (37.72 words), suggesting that longer mixed-sentiment reviews provide conflicting lexical evidence that confounds the bag-of-words representation.

Effect of Class Weighting

Applying `class_weight='balanced'` produced markedly different effects across models, as shown in Figure 6. Logistic Regression and Linear SVM both improved substantially in Neutral F1, while Complement Naive Bayes showed no change due to the lack of support for this parameter. These results highlight that the ability to respond to class imbalance mitigation is itself model-dependent, and that any improvement in minority-class F1 comes at some cost to overall accuracy — a tradeoff that must be weighed against application requirements.

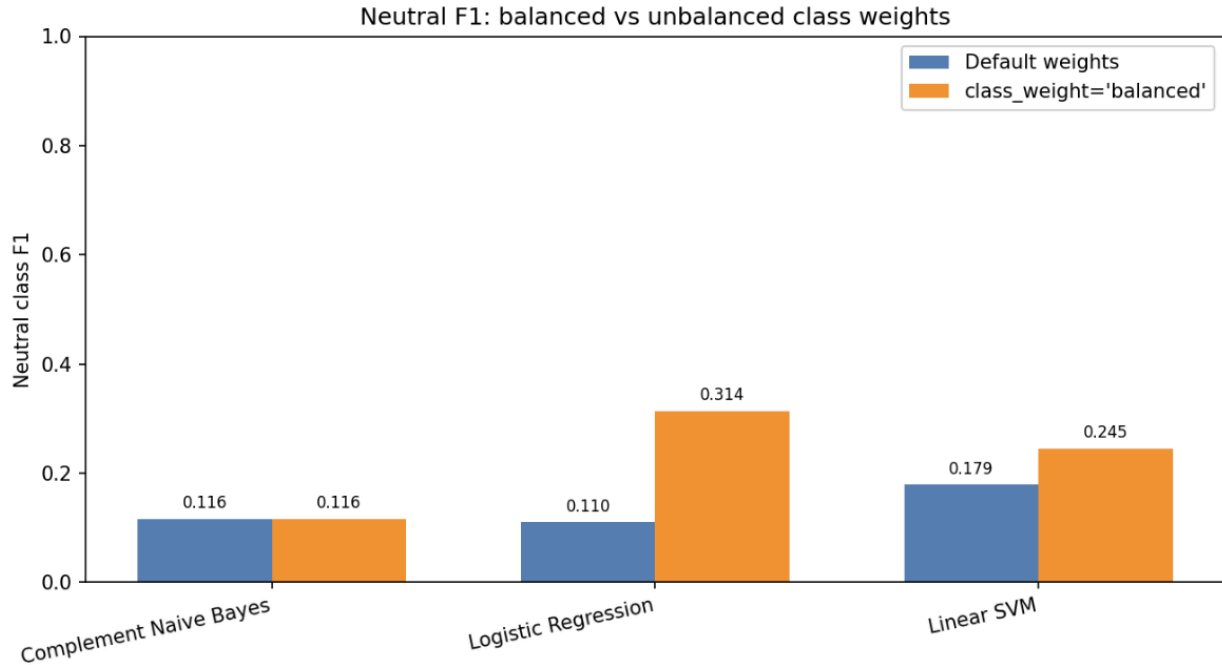


Figure 6. Neutral class F1 score comparison between default and class-weighted training for all three models.

Conclusions

This project demonstrates that traditional machine learning methods combined with TF-IDF features can achieve strong performance on binary sentiment classification of product reviews, with all three models reaching approximately 87–88% accuracy on the All Beauty category. However, the introduction of a Neutral class exposes two structural limitations of this approach.

The first limitation is sensitivity to class imbalance. When Neutral reviews account for only 8% of the data, all three models default to largely ignoring this class, achieving near-zero recall despite reasonable overall accuracy. Class weighting is a simple and effective mitigation for models that support it, but even the best balanced model's Neutral F1 remained far below the performance on the other two classes, as shown in Figure 6. This finding is consistent with the broader observation in Devika et al. (2016) that machine learning classifiers often struggle to generalize across skewed distributions.

The second limitation is the inadequacy of bag-of-words representations for capturing mixed or ambiguous sentiment. TF-IDF treats each word independently, making it unable to represent the way that Neutral reviews often balance positive and negative language within the same sentence. The word count finding — correctly classified Neutral reviews averaged 31.78 words versus 37.72 for misclassified ones — reinforces this: longer, more mixed reviews provide conflicting lexical evidence that confounds the representation. Birjali et al. (2021) advocate for

transformer-based models such as BERT precisely because they encode full sequence context rather than token frequencies in isolation, a capability that would be expected to substantially improve Neutral class identification.

The comparison between accuracy and macro F1 also carries a methodological lesson applicable beyond this project. In any real-world application where minority-class performance matters — for example, a system that monitors for ambiguous customer feedback — relying on accuracy alone would lead to systematically poor decisions.

Future work could extend this project in several directions. First, applying fine-tuned language models would provide a meaningful upper bound on this task and clarify how much of the Neutral classification failure is due to the feature representation versus the training objective. Second, training on one product category and testing on another would assess domain generalization, directly addressing the concern raised by Devika et al. (2016) about cross-domain performance. Third, alternative label definitions — such as treating 2–4 star reviews as a broader mixed class — might produce a better-balanced dataset that reduces the structural difficulty of neutral detection.

Works Cited

Birjali, M., Kasri, M., & Beni-Hssane, A. (2021). A comprehensive survey on sentiment analysis: Approaches, challenges and trends. *Knowledge-Based Systems*, 226, 107134. <https://doi.org/10.1016/j.knosys.2021.107134>

Devika, M. D., Sunitha, C., & Ganesh, A. (2016). Sentiment analysis: A comparative study on different approaches. *Procedia Computer Science*, 87, 44–49. <https://doi.org/10.1016/j.procs.2016.05.124>

Hou, Y., Li, J., He, Z., Yan, A., Chen, X., & McAuley, J. (2024). Bridging language and items for retrieval and recommendation. *arXiv preprint arXiv:2403.03952*.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., ... & Duchesnay, E. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.

Rennie, J. D., Shih, L., Teevan, J., & Karger, D. R. (2003). Tackling the poor assumptions of naive Bayes text classifiers. In *Proceedings of the 20th International Conference on Machine Learning (ICML-03)* (pp. 616–623).

Appendix

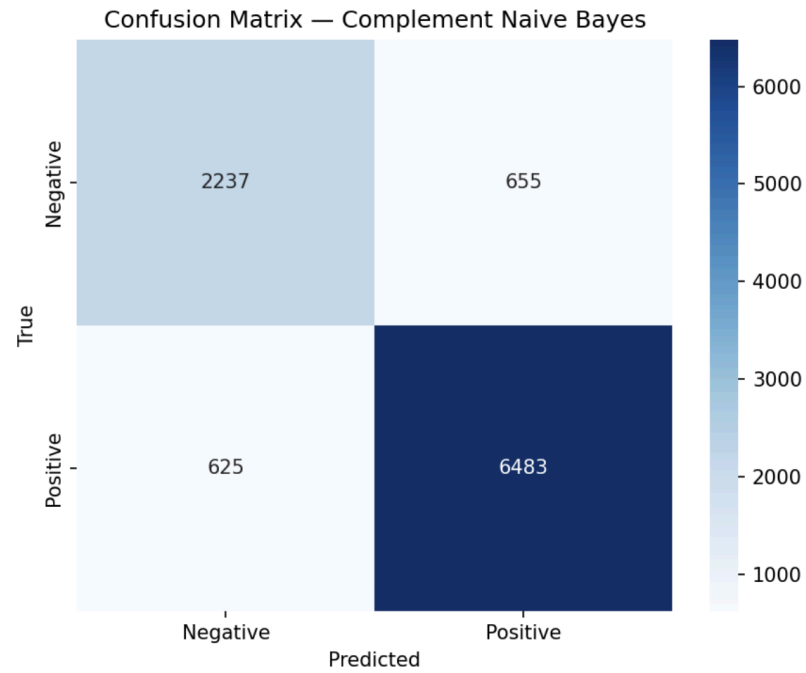


Figure A1. Confusion matrix for Complement Naive Bayes — binary classification.

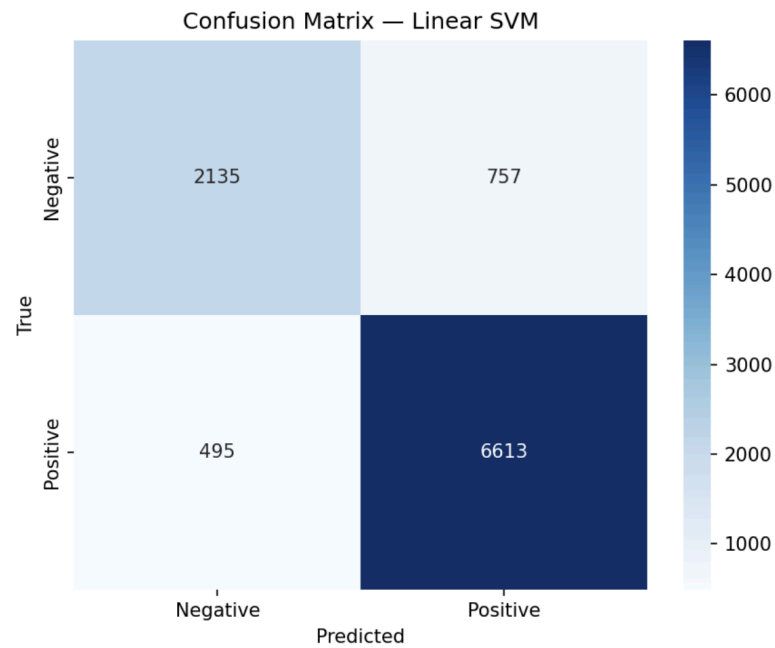


Figure A2. Confusion matrix for Linear SVM — binary classification.

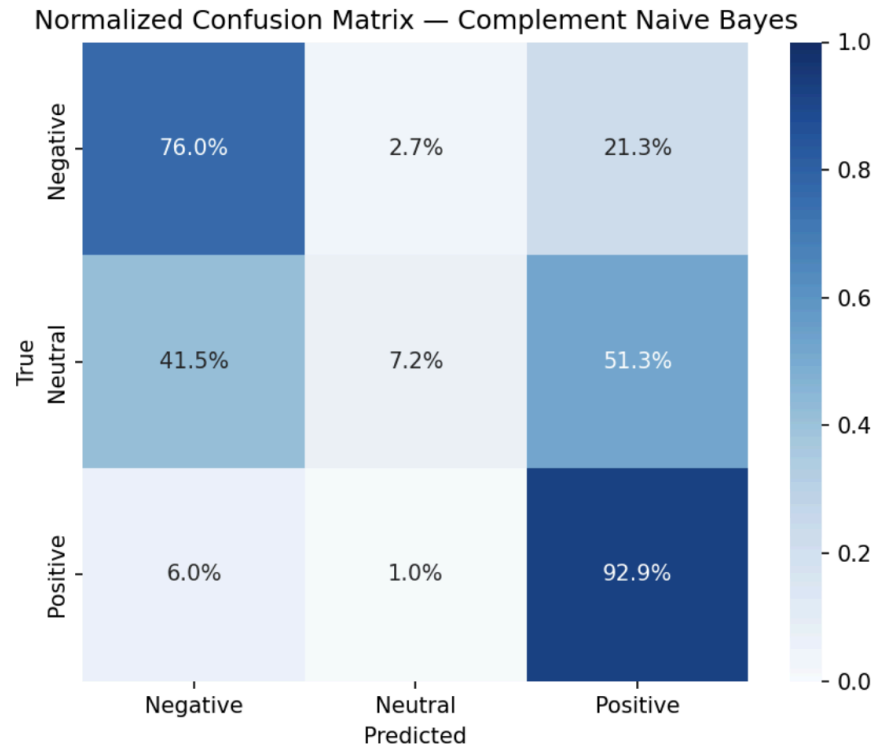


Figure A3. Normalized confusion matrix for Complement Naive Bayes — ternary classification.

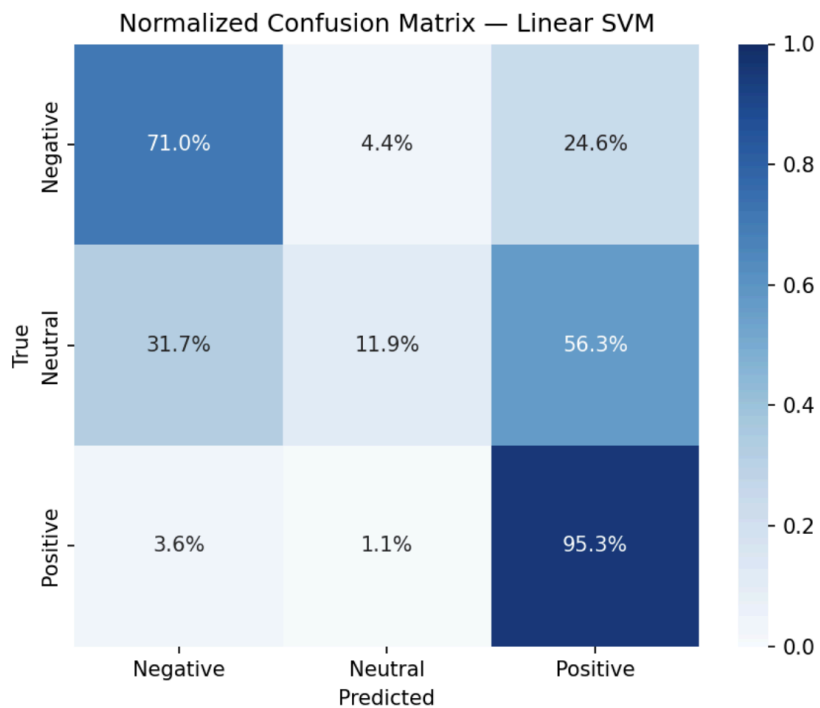


Figure A4. Normalized confusion matrix for Linear SVM — ternary classification.



Figure A5. Misclassification breakdown by error type for Logistic Regression — ternary classification.